ANFIS Based Student Performance Prediction System

1. Introduction

Artificial Intelligence techniques such as fuzzy logic and neural networks are widely used for prediction tasks. ANFIS (Adaptive Neuro-Fuzzy Inference System) combines both approaches to create an intelligent system capable of learning from data.

In this project, ANFIS is used to predict student performance based on academic inputs.

2. Objective

The objectives of this experiment are:

- To design an ANFIS model for student performance prediction
- To integrate fuzzy logic with neural networks
- To train the model using a dataset
- To evaluate model performance using RMSE

3. Inputs and Output

Inputs:

- Attendance (0–100)
- Assignment Marks (0–100)
- Test Marks (0–100)

Output:

- Performance Score
- Performance Level (Poor / Average / Good)

4. Dataset

A dataset of 1000 student records is used.

Each record contains:

- Attendance
- Assignment marks
- Test marks
- Performance score

5. Integration of Fuzzy Logic and Neural Networks

Fuzzy logic provides rule-based reasoning using membership functions. ANFIS integrates neural networks to automatically adjust these membership functions and rules based on data.

This combination improves prediction accuracy and enables the system to learn from examples.

6. Methodology

The following steps are used:

- Dataset normalized to range 0–1
- Data split into training (80%) and testing (20%)
- Initial FIS generated using membership functions
- ANFIS model trained over multiple epochs
- Validation used to monitor performance
- Early stopping applied to avoid overfitting

7. Results

The trained ANFIS model produced the following results:

- Training RMSE = 0.0311324
- Checking RMSE = 0.0323606

Lower RMSE values indicate better prediction accuracy.

- Best Epoch = 1
- Final Test RMSE ≈ 3.24

The model shows minimal difference between training and checking error, indicating good generalization.

8. Output Example

Input:

- Attendance = 85
- Assignment = 80
- Test = 78

Output:

- Predicted Score = 80.09
- Performance Level = Good

9. Training Process

The ANFIS training process includes initialization of parameters and iterative optimization over epochs to minimize error.

10. Screenshots

The following figures illustrate the ANFIS training process, error convergence, and final prediction results:

- Training vs Testing Error Graph

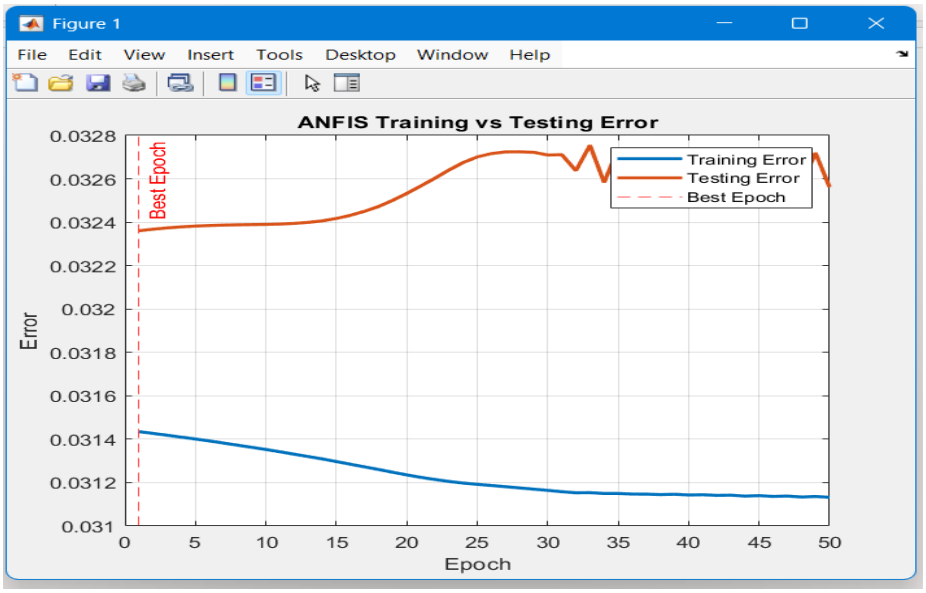

Fig1: Training VS Testing Error Graph

- Training Process (ANFIS info + epochs)

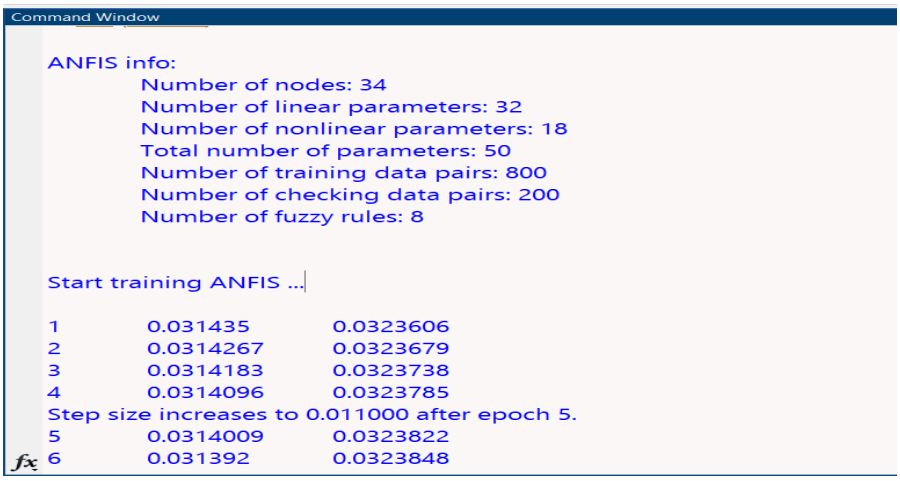

Fig2: Training Process

- Final Output

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Minimal training RMSE = 0.0311324
Minimal checking RMSE = 0.0323606
Best Epoch = 1
Final Test RMSE = 3.24

Predicted Score = 80.09
Performance Level = Good
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Fig3: Final Output

11. Conclusion

The ANFIS model successfully predicts student performance with good accuracy. The combination of fuzzy logic and neural networks enhances adaptability and learning capability of the system.